

# JAVASCRIPT EXPERIMENTS

## Experiment 1: Display Message using JavaScript

### Aim

To display a message using JavaScript.

### Algorithm

1. Start the HTML document.
2. Add a <script> section inside the <head> or <body>.
3. Use document.write() to display a message.
4. Save the file with .html extension.
5. Open the file in a web browser.

### Code

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>JavaScript Example</title>
```

```
</head>
```

```
<body>
```

```
<script>
```

```
document.write("Welcome to JavaScript");
```

```
</script>
```

```
</body>
```

```
</html>
```

### Output

The webpage displays the message:

## Welcome to JavaScript

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### Experiment 2: Display Alert Message

#### Aim

To display a popup alert message using JavaScript.

#### Algorithm

1. Create an HTML page.
2. Add a <script> section.
3. Use alert() function to display a popup message.
4. Save and run the program.

#### Code

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<script>
```

```
alert("Welcome to Web Programming");
```

```
</script>
```

```
</body>
```

```
</html>
```

#### Output

A **popup alert box** displaying:

**Welcome to Web Programming**

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### Experiment 3: Addition of Two Numbers

#### Aim

To perform addition of two numbers using JavaScript.

### Algorithm

1. Create an HTML page.
2. Declare two variables in JavaScript.
3. Store values in the variables.
4. Add the two numbers.
5. Display the result using document.write().

### Code

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<script>
```

```
var a = 10;
```

```
var b = 20;
```

```
var sum = a + b;
```

```
document.write("Sum = " + sum);
```

```
</script>
```

```
</body>
```

```
</html>
```

### Output

The webpage displays:

**Sum = 30**

---

### Experiment 4: JavaScript Button Click Event

#### Aim

To display a message when a button is clicked.

### Algorithm

1. Create an HTML page.
2. Add a button element.
3. Create a JavaScript function.
4. Call the function when the button is clicked using onclick.
5. Display a message using alert().

### Code

```
<!DOCTYPE html>

<html>


<body>


<button onclick="showMessage()">Click Me</button>


<script>

function showMessage()
{
alert("Button Clicked!");
}

</script>

</body>

</html>
```

### Output

When the button is clicked, an **alert message appears**.

---

## Experiment 5: Change Text using JavaScript

### Aim

To change webpage text dynamically using JavaScript.

### Algorithm

1. Create an HTML page.
2. Add a paragraph with an ID.
3. Create a button.
4. Use JavaScript getElementById() method.
5. Change the paragraph text when the button is clicked.

### Code

```
<!DOCTYPE html>

<html>


<body>


<p id="demo">Original Text</p>


<button onclick="changeText()">Change Text</button>


<script>


function changeText()
{
document.getElementById("demo").innerHTML="Text Changed!";
}


</script>


</body>
</html>
```

### Output

Clicking the button **changes the paragraph text**.

---

## Experiment 6: Form Validation using JavaScript

### Aim

To validate form input using JavaScript.

### Algorithm

1. Create an HTML form.
2. Add a text input field.
3. Create a JavaScript validation function.
4. Check whether the field is empty.
5. Display an alert if empty.
6. Prevent form submission.

### Code

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<form onsubmit="return validate()">
```

Name:

```
<input type="text" id="name">
```

```
<input type="submit">
```

```
</form>
```

```
<script>
```

```
function validate()
```

```
{
```

```
var name=document.getElementById("name").value;
```

```
if(name=="")
{
alert("Name cannot be empty");
return false;
}

}
```

```
</script>
```

```
</body>
```

```
</html>
```

### Output

If the user submits the form **without entering a name**, an alert message appears.

---

## Experiment 7: JavaScript Calculator

### Aim

To create a simple calculator using JavaScript.

### Algorithm

1. Create two input fields.
2. Create a button for addition.
3. Use JavaScript function to get input values.
4. Convert values to numbers.
5. Display the result.

### Code

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

Number 1:

```
<input type="number" id="num1">
```

Number 2:

```
<input type="number" id="num2">
```

```
<button onclick="add()">Add</button>
```

```
<p id="result"></p>
```

```
<script>
```

```
function add()
```

```
{
```

```
var a=parseInt(document.getElementById("num1").value);
```

```
var b=parseInt(document.getElementById("num2").value);
```

```
var sum=a+b;
```

```
document.getElementById("result").innerHTML="Result = "+sum;
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

### Output

Displays the **sum of two numbers**.

---

## Experiment 8: Image Slider (Previous/Next)

### Aim

To create an image slider using JavaScript.

### Algorithm

1. Create an image element.
2. Store image names in an array.
3. Create Next and Previous buttons.
4. Use JavaScript functions to change images.
5. Update image source dynamically.

### Code

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```

```

```
<br><br>
```

```
<button onclick="prev()">Previous</button>
```

```
<button onclick="next()">Next</button>
```

```
<script>
```

```
var images=["img1.jpg","img2.jpg","img3.jpg"];
```

```
var i=0;
```

```
function next()
```

```
{
```

```
  i++;
```

```
if(i>=images.length)
{
i=0;
}

document.getElementById("slider").src=images[i];
}

function prev()
{
i--;

if(i<0)
{
i=images.length-1;
}

document.getElementById("slider").src=images[i];
}

</script>
```

</body>

</html>

### **Output**

User can **navigate images using Previous and Next buttons.**

---

## **Experiment 9: Digital Clock**

### **Aim**

To create a digital clock using JavaScript.

### **Algorithm**

1. Create an HTML element to display time.
2. Use JavaScript Date() object.
3. Extract current time.
4. Update time every second using setInterval().
5. Display the time.

#### **Code**

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2 id="clock"></h2>
```

```
<script>
```

```
setInterval(showTime,1000);
```

```
function showTime()
```

```
{
```

```
var d=new Date();
```

```
var time=d.toLocaleTimeString();
```

```
document.getElementById("clock").innerHTML=time;
```

```
}
```

```
</script>
```

```
</body>
```

</html>

### Output

A **live digital clock** showing current time.

---

## Experiment 10: Change Background Color

### Aim

To change webpage background color using JavaScript.

### Algorithm

1. Create an HTML page.
2. Add a button element.
3. Create a JavaScript function.
4. Change background color using document.body.style.
5. Execute the function on button click.

### Code

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<button onclick="changeColor()">Change Color</button>
```

```
<script>
```

```
function changeColor()
```

```
{
```

```
document.body.style.backgroundColor="lightgreen";
```

```
}
```

```
</script>
```

```
</body>
```

</html>

## Output

Clicking the button **changes the background color of the webpage.**

---

## Final Experiment: Student Registration Form with JavaScript Validation

### Aim

To design a **Student Registration Form** and perform **form validation using JavaScript** to ensure that required fields are properly filled before submission.

---

### Algorithm

1. Start the HTML document using <!DOCTYPE html>.
  2. Create the <html>, <head>, and <title> sections.
  3. Add a <script> section to write the JavaScript validation function.
  4. Create a function named validateForm().
  5. Inside the function, retrieve form values using document.getElementById().
  6. Check whether **Name field is empty**.
  7. Check whether **Email field is empty**.
  8. Check whether **Password length is less than 6 characters**.
  9. If any field is invalid, display an **alert message**.
  10. Prevent form submission using return false.
  11. Create the registration form using <form>.
  12. Add input fields for **Name, Email, Password, Gender, and Course**.
  13. Attach the validation function using onsubmit.
  14. Save the file with .html extension.
  15. Run the file in a web browser.
- 

### Code

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Student Registration Form</title>
```

```
<script>
```

```
function validateForm()
```

```
{
```

```
var name = document.getElementById("name").value;
```

```
var email = document.getElementById("email").value;
```

```
var password = document.getElementById("password").value;
```

```
if(name == "")
```

```
{
```

```
alert("Name cannot be empty");
```

```
return false;
```

```
}
```

```
if(email == "")
```

```
{
```

```
alert("Email cannot be empty");
```

```
return false;
```

```
}
```

```
if(password.length < 6)
```

```
{
```

```
alert("Password must be at least 6 characters");
```

```
return false;
```

```
}
```

```
alert("Registration Successful");
```

```
return true;
```

```
}
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<h2>Student Registration Form</h2>
```

```
<form onsubmit="return validateForm()">
```

```
Name:<br>
```

```
<input type="text" id="name"><br><br>
```

```
Email:<br>
```

```
<input type="email" id="email"><br><br>
```

```
Password:<br>
```

```
<input type="password" id="password"><br><br>
```

```
Gender:<br>
```

```
<input type="radio" name="gender"> Male
```

```
<input type="radio" name="gender"> Female
```

```
<br><br>
```

```
Course:<br>
```

```
<select>
<option>B.Tech</option>
<option>M.Tech</option>
<option>BCA</option>
<option>MCA</option>
</select>

<br><br>

<input type="submit" value="Register">

</form>

</body>
</html>
```

---

## Output

The webpage displays a **Student Registration Form** containing:

- Name field
- Email field
- Password field
- Gender selection
- Course dropdown
- Register button

## Validation Behavior

If the user:

- Leaves **Name empty** → Alert: *Name cannot be empty*
- Leaves **Email empty** → Alert: *Email cannot be empty*
- Enters **password less than 6 characters** → Alert: *Password must be at least 6 characters*

If all fields are correct:

**Registration Successful** message appears.